A UNIVERSALLY CLOSED MORPHISM OF SCHEMES IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-universally-closed-quasi-compact is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 9985-9991 and proof span 9993-10047 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains a reference environment attributing the result to Bjorn Poonen; the project's own shared preamble (retained in full at 88 lines) defines that environment as a comment environment, so the attribution is not typeset, exactly as in the project's own output, and no bibliography is attached.

Editorial note. The proof names an auxiliary affine space with one coordinate per affine open of the source and writes the corresponding closed subset explicitly; the author's notation is preserved unchanged.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The characterization of quasi-compactness by the existence of an affine open whose preimage is not quasi-compact, cited from the schemes chapter. Cited at proof lines 9995-9997 and again at 10011-10013. Produces the affine base to work over and the bad point at which every neighbourhood has non-quasi-compact preimage.
- The description of points of a fibre product by a pair of points together with a prime in the tensor product of their residue fields, cited from the schemes chapter. Cited at proof lines 10038-10040. Allows the point constructed over a chosen point of X and a chosen point of the auxiliary affine space to be lifted to the fibre product.
- The description of the closed subscheme cut out by a family of coordinate functions as a polynomial algebra on the remaining coordinates, and the bijection between its points over a given base point and the spectrum of the same polynomial algebra over that residue field. Used at proof lines 10033-10037. Makes it possible to choose the auxiliary point avoiding the

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zero locus of a prescribed polynomial while lying in the vanishing loci of the omitted coordinates.

Source reference. Original lemma 01JT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2026–2026. Same-chapter reference used in the retained author text at source line(s) 1326, 1485, 1652, 3643, 10040. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01K4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2037–2037. Same-chapter reference used in the retained author text at source line(s) 1727, 1761, 2213, 2252, 9996, 10014. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *A universally closed morphism of schemes is quasi-compact.*

Proof. Let $f : X \to S$ be a morphism. Assume that f is not quasi-compact. Our goal is to show that f is not universally closed. By Schemes, Lemma 01K4 there exists an affine open $V \subset S$ such that $f^{-1}(V)$ is not quasi-compact. To achieve our goal it suffices to show that $f^{-1}(V) \to V$ is not universally closed, hence we may assume that $S = \mathrm{Spec}(A)$ for some ring A.

Write $X = \bigcup_{i \in I} X_i$ where the X_i are affine open subschemes of X. Let $T = \mathrm{Spec}(A[y_i; i \in I])$. Let $T_i = D(y_i) \subset T$. Let Z be the closed set $(X \times_S T) - \bigcup_{i \in I}(X_i \times_S T_i)$. It suffices to prove that the image $f_T(Z)$ of Z under $f_T : X \times_S T \to T$ is not closed.

There exists a point $s \in S$ such that there is no neighborhood U of s in S such that X_U is quasi-compact. Otherwise we could cover S with finitely many such U and Schemes, Lemma 01K4 would imply f quasi-compact. Fix such an $s \in S$.

First we check that $f_T(Z_s) \neq T_s$. Let $t \in T$ be the point lying over s with $\kappa(t) = \kappa(s)$ such that $y_i = 1$ in $\kappa(t)$ for all i. Then $t \in T_i$ for all i, and the fiber of $Z_s \to T_s$ above t is isomorphic to $(X - \bigcup_{i \in I} X_i)_s$, which is empty. Thus $t \in T_s - f_T(Z_s)$.

Assume $f_T(Z)$ is closed in T. Then there exists an element $g \in A[y_i; i \in I]$ with $f_T(Z) \subset V(g)$ but $t \notin V(g)$. Hence the image of g in $\kappa(t)$ is nonzero. In particular some coefficient of g has nonzero image in $\kappa(s)$. Hence this coefficient is invertible on some neighborhood U of s. Let J be the finite set of $j \in I$ such that y_j appears in g. Since X_U is not quasi-compact, we may choose a point $x \in X - \bigcup_{j \in J} X_j$ lying above some $u \in U$. Since g has a coefficient that is invertible on U, we can find a point $t' \in T$ lying above u such that $t' \notin V(g)$ and $t' \in V(y_i)$ for all $i \notin J$. This is true because $V(y_i; i \in I, i \notin J) = \mathrm{Spec}(A[y_j; j \in J])$ and the set of points of this scheme lying over u is bijective with $\mathrm{Spec}(\kappa(u)[y_j; j \in J])$. In other words $t' \notin T_i$ for each $i \notin J$. By Schemes, Lemma 01JT we can find a point z of $X \times_S T$ mapping to $x \in X$ and to $t' \in T$. Since $x \notin X_j$ for $j \in J$ and $t' \notin T_i$ for $i \in I \setminus J$ we see that $z \in Z$. On the other hand $f_T(z) = t' \notin V(g)$ which contradicts $f_T(Z) \subset V(g)$. Thus the assumption "$f_T(Z)$ closed" is wrong and we conclude indeed that f_T is not closed, as desired. $\square$